

STUDENT PERFORMANCE PREDICTION & ACADEMIC ANALYSIS

Using Multiple Linear Regression

Integrated MTech AI Project
VIT Bhopal University

Terminal-Based AI/ML Application

Project Report

1. Introduction

This project presents a lightweight terminal-based AI/ML application for student academic analysis. It combines structured academic indicators with Multiple Linear Regression to estimate a student's final performance. A separate academic-analysis layer converts raw assessment values into weighted contributions and checks the defined eligibility and pass conditions used by the project.

2. Problem Statement

Academic information may be distributed across study habits, attendance, continuous assessment, internal tests and final examination performance. A single workflow can make these indicators easier to interpret by producing a model-based estimate together with transparent academic calculations and recommendations.

3. Objectives

The objectives are to build an executable AI/ML project using Multiple Linear Regression; process a structured academic dataset; train and test the model using an 80/20 split; evaluate it using MAE, RMSE and R^2 ; accept individual student inputs; calculate academic contributions; check defined conditions; and produce understandable recommendations.

4. Functional Requirements

The system loads the CSV dataset, validates input ranges, trains Multiple Linear Regression, predicts a final score, calculates CAM/attendance/CAT-1/CAT-2/TEE contributions, checks defined attendance/TEE/raw-score conditions, generates recommendations, and supports interactive, example, metrics and validation-test command-line modes.

5. Non-Functional Requirements

Usability is provided through simple terminal prompts and structured output. Reliability is supported by input validation and automated tests. Maintainability is improved through separated Python modules. Resource efficiency is supported by a standard-library-only implementation. Portability is supported for Python 3.9+ on common desktop operating systems. Error handling rejects invalid numerical input with a clear message.

6. System Architecture

The application is divided into configuration, data loading, regression, academic calculation, validation/recommendation, reporting and application orchestration modules. The command-line entry point in main.py coordinates these modules.

7. Project Workflow

The workflow is: user input → validation → dataset loading → 80/20 split → Multiple Linear Regression training → prediction → academic calculation → eligibility/pass analysis → recommendation → model evaluation and terminal output.

8. Dataset

The supplied repository contains 240 simulated academic records. Features include study hours, attendance, CAM percentage, previous marks, CAT-1 raw marks, CAT-2 raw marks and TEE raw marks. The target is Final_Score_100. The dataset is synthetic and is included only for educational demonstration; it is not real VIT student data.

9. Machine Learning Method

The model is ordinary least-squares Multiple Linear Regression. The implementation solves the normal-equation system using Gauss-Jordan elimination with partial pivoting and does not require external machine-learning libraries. The seven predictors are study hours, attendance, CAM percentage, previous marks, CAT-1, CAT-2 and TEE.

10. Train-Test Design

The 240 records are divided using an 80/20 split: 192 records for training and 48 records for testing. The model is fitted only on the training records, and evaluation is performed on the held-out test records.

11. Model Evaluation

The observed evaluation results are approximately: MAE 1.13, RMSE 1.32, R^2 0.9831 and baseline MAE 8.86. These values demonstrate the implemented evaluation workflow. Because the dataset is simulated and the target is strongly related to the predictor variables, the high R^2 should not be interpreted as real-world predictive accuracy.

12. Academic Analysis

The application converts raw assessment values into weighted contributions, produces an internal total and TEE contribution, calculates a final score, and checks the project-defined attendance, TEE and raw-score conditions. The output clearly distinguishes the ML prediction from the calculated academic score.

13. Implementation

The source is modularized into config.py, data_loader.py, regression.py, academic.py, analysis.py, reporting.py and app.py, with main.py acting as the command-line entry point. This separation keeps data handling, model computation, academic logic and presentation independently understandable.

14. Testing

Automated validation checks the 240-record dataset, 80/20 split, regression metrics, baseline comparison and academic calculations. Manual execution was also performed for the interactive mode, the example mode and the metrics mode in Windows PowerShell. The project returned successful results without runtime errors.

15. Limitations

The dataset is synthetic rather than institutional. The model is not validated on real VIT student records. Predictions are prototype estimates and are not official university results. The application does not collect university credentials or connect to a university portal.

16. Future Enhancements

Future work could include a larger validated dataset, cross-validation, comparison with additional regression models, feature explainability, persistent student profiles, a database layer, subject-wise analytics and authorized institutional API integration where appropriate.

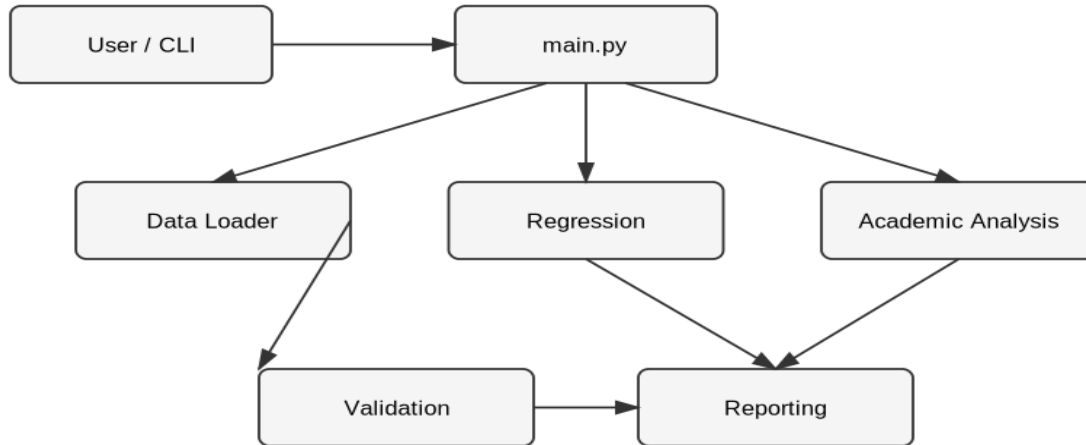
17. Conclusion

The project demonstrates an end-to-end introductory AI/ML workflow in a command-line executable form. It integrates data processing, regression, evaluation, prediction, academic calculations and interpretable recommendations in a modular and resource-efficient implementation.

18. Architecture and UML-Style Diagrams

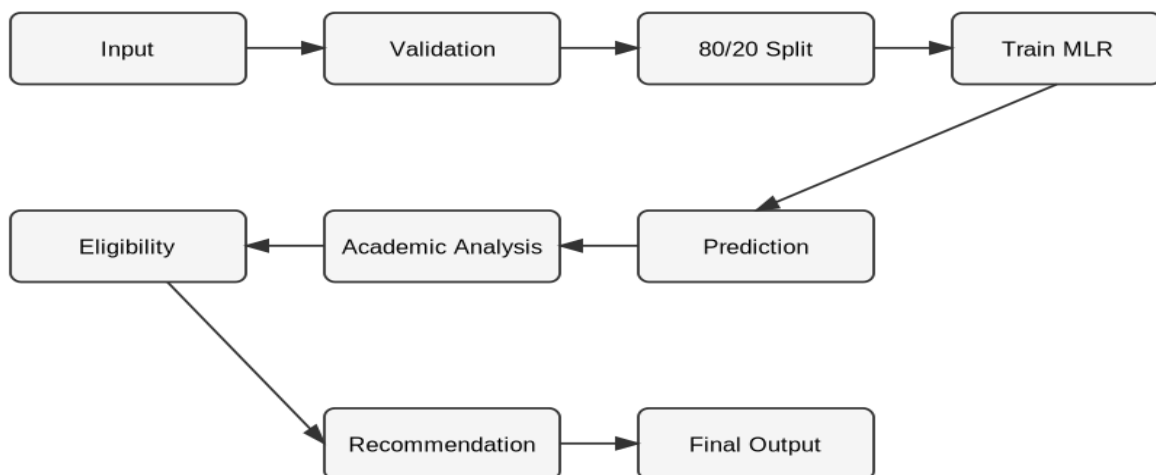
System Architecture

System Architecture



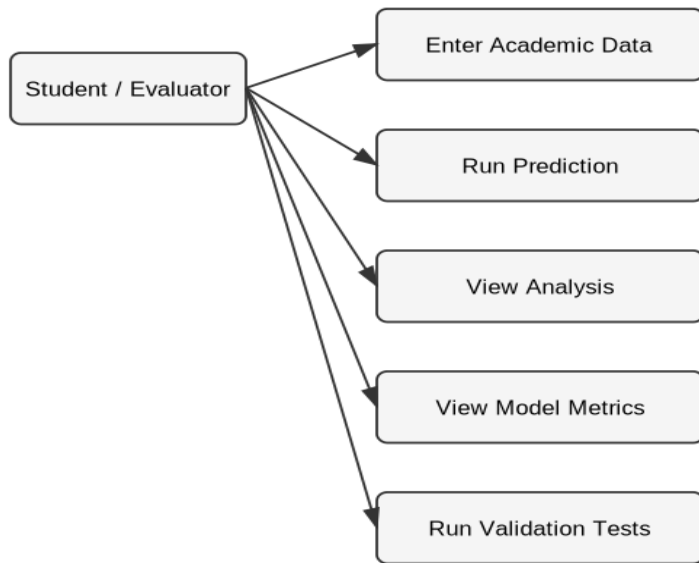
Workflow

Project Workflow



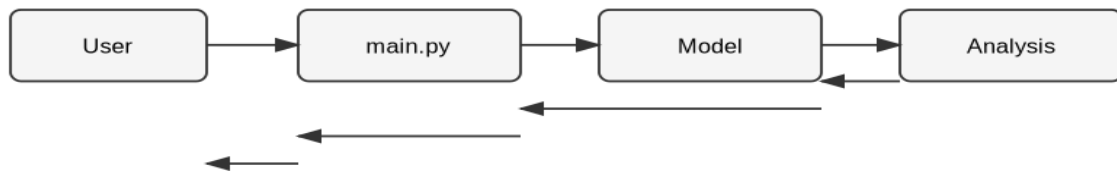
Use Case Diagram

Use Case Diagram



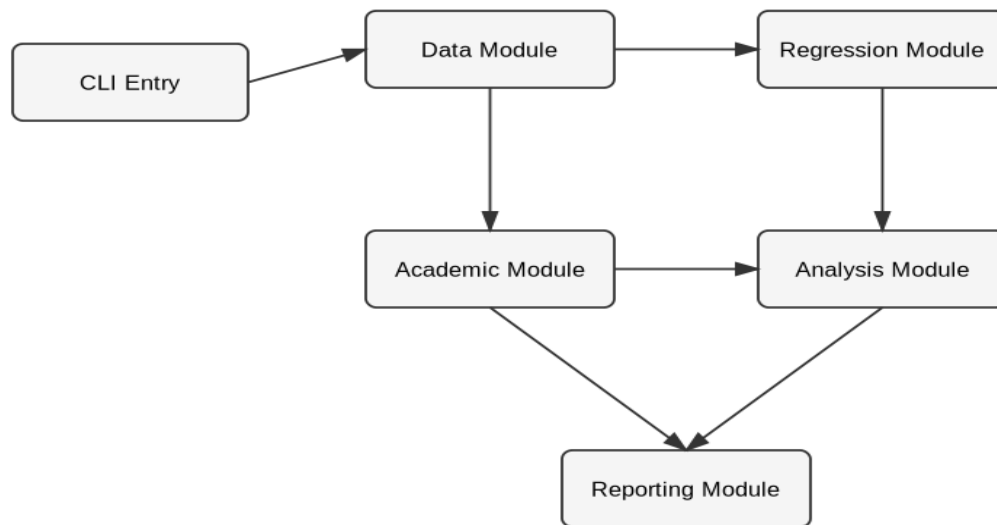
Sequence Diagram

Sequence Diagram



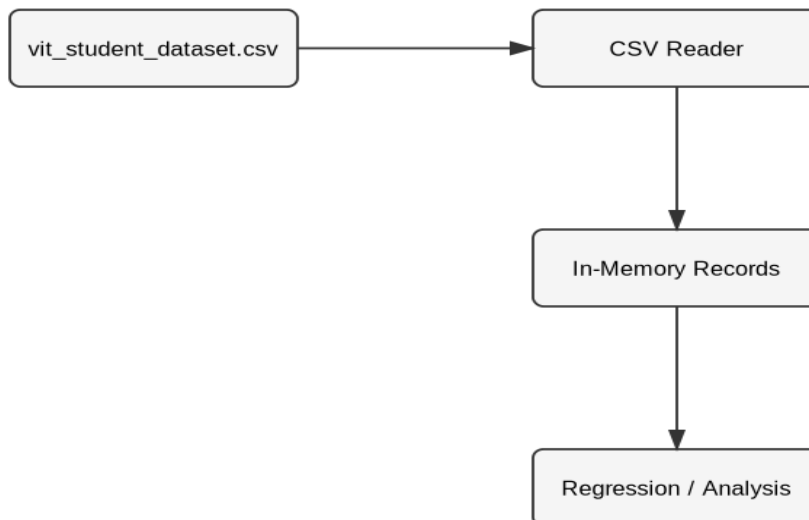
Component Diagram

Component Diagram



Storage Design

Storage Design



19. Execution Evidence

The following screenshots document successful local execution of the terminal application during project verification.

```
All CLI validation tests passed.
Records: 240 | Test MAE: 1.13 | RMSE: 1.32 | R2: 0.9831
PS C:\Users\Shlok\Downloads\VIT_Student_Performance_Predictor_TE
RMINAL_READY> python main.py --test
All CLI validation tests passed.
Records: 240 | Test MAE: 1.13 | RMSE: 1.32 | R2: 0.9831
PS C:\Users\Shlok\Downloads\VIT_Student_Performance_Predictor_TE
RMINAL_READY> |
```

```
STUDENT PERFORMANCE ANALYSIS
-----
ML predicted final score : 76.55/100
Performance level       : Good
CAM contribution        : 27.30/35
Attendance contribution  : 5.00/5
CAT-1 contribution      : 11.40/15
CAT-2 contribution      : 12.00/15
Internal total          : 55.70/70
TEE contribution        : 21.60/30
Calculated final score  : 77.30/100

ELIGIBILITY / PASS CHECK
-----
Attendance >= 75%       : Eligible
TEE >= 40/100           : Satisfied
CAT1+CAT2+TEE >= 80/200 : Satisfied
Overall pass condition   : PASS

RECOMMENDATION
-----
- Maintain the current level and use timed revision/practice.
- Use the ML prediction as a prototype estimate, not an official
  university result.

MODEL EVALUATION
-----
Dataset records : 240
Train records   : 192
Test records    : 48
MAE              : 1.13
RMSE             : 1.32
R2               : 0.9831
```

```

Note: the supplied dataset is simulated for educational demonstration.
PS C:\Users\Shlok\Downloads\VIT_Student_Performance_Predictor_TERMINAL_READY> python main.py
=====
VIT STUDENT PERFORMANCE PREDICTION & ACADEMIC ANALYSIS
Terminal Edition
=====

Loaded 240 simulated records.
Multiple Linear Regression trained using an 80/20 split.

Enter student data.
Study hours per day (0-12): 5
Attendance % (0-100): 85
CAM % (0-100): 78
Previous final score (0-100): 72
CAT-1 raw marks (0-50): 38
CAT-2 raw marks (0-50): 40
TEE raw marks (0-100): 72

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MODEL EVALUATION
-----
Dataset records : 240

```

20. Command-Line Execution

```
python main.py
python main.py --test
python main.py --metrics
python main.py --example
python main.py --study 5 --attendance 86 --cam 78 --previous 72 --cat1 38 --cat2 40 --tee 72
```

21. References

VIT Bhopal University, provided academic regulation document used as the project reference for the assessment structure represented by the prototype.

VIT Bhopal University, provided Curriculum document identifying Fundamentals in AI and ML as a programme-core course and its learning objectives.

Project dataset and source code contained in this repository.